



UNIVERSITY CATEGORY

REGISTRATION FORM ROUND 1

BACH KHOA INNOVATION 2026

1. TEAM'S NAME: MORPHYSICS

2. TEAM LEADER

Full name: TÔN LONG QUÂN

ID Number (if any): 2554035

Email (HCMUT/ personal email): quan.ton091207@hcmut.edu.vn

Phone number: 0964474527

3. MENTOR (IF ANY)

Full name: TRƯƠNG THỊ THÁI MINH

Title: Doctor

Working unit: Faculty of Computer Science and Engineering

Email: thaiminh@hcmut.edu.vn

Phone number: 0943777414

PROJECT'S INFORMATION

1. PROJECT'S NAME: MORPHYSICS

- *Applied Sciences;*
- *Computer Science and Engineering;*

2. SUMMARIZING YOUR IDEA/PROJECT

Science education in Vietnam is currently facing a major gap between the new practical curriculum and the lack of actual laboratory equipment. As the national education system shifts toward real-world applications, practical problems now make up a large portion of the high school curriculum. However, schools only meet about 54.3% of the required equipment standards. Because of this shortage, combined with large class sizes and short class periods, teachers are forced to rely on "dry teaching" methods where students just listen and take notes. As a result, students have to memorize formulas to pass exams without truly understanding how things work, leading to frustration and a drop in students choosing STEM subjects.

To solve this lack of hands-on learning, Morphysics was developed, an AI-powered web application that automatically turns textbook problems into interactive virtual labs. Instead of making users build digital setups by hand, Morphysics uses a smart Text-to-Simulation feature. The AI can extract physical details from typed texts of physics problems. It then creates a custom, 2D simulation running in real-time for students to gain a deep, comprehensive understanding of the problem.

This automated system provides a safe and engaging space where students can actively experiment with physics. Inside the virtual lab, learners can easily drag objects, change numbers, or test extreme limits to see what happens immediately. They get to learn through trial and error without the fear of breaking expensive real-world tools or facing any physical safety risks.

Ultimately, Morphysics brings clear benefits to everyone in the educational system without requiring numerous new physical hardware. For students, it replaces boring memorization with engaging visual learning, helping them truly master the subject. For teachers, it acts as a fast



teaching assistant that generates visual aids in seconds, perfectly matching the practical needs of the new curriculum.

3. DETAILS OF YOUR IDEA/PROJECT

Morphysics is an innovative, AI-powered interactive physics laboratory web application designed to revolutionize STEM education. The platform serves as a dynamic bridge between abstract theoretical physics problems and practical, visual experimentation. By implementing advanced artificial intelligence, Morphysics extract variables from user inputs and instantly transform them into highly accurate, customized 2D interactive simulations.

A significant discrepancy exists in contemporary science education: students are thoroughly exposed to scientific theories but severely lack hands-on laboratory experience. Schools face immense financial and logistical burdens to maintain physical labs, and educators are constrained by short 45-minute class periods, forcing a reliance on “dry teaching”. Furthermore, fear of damaging expensive equipment and strict safety protocols frequently reduce students to passive observers. To address these challenges, Morphysics provides a clear, scalable, and highly appropriate solution to these barriers. As a zero-setup virtual canvas powered by a precise physics engine, it entirely bypasses the steep costs, maintenance requirements, and safety liabilities of traditional physical labs. It grants every student an autonomous, safe sandbox environment where they can freely manipulate variables, such as dropping heavy weights, adjusting friction, or short-circuiting electrical wires to observe immediate consequences without any real-world danger.

What truly makes the product innovative and unique is its creative approach to physics visualization, directly addressing challenges with custom solutions that outperform existing educational alternatives. To effectively visualize physics, a tool must be accurate, adaptable, and interactive. While traditional step-by-step solvers output mathematical steps from scanned images, Morphysics translates that same mathematical data into an interactive visual simulation. Furthermore, unlike existing manual simulation platforms that rely on pre-built, static scenarios, Morphysics dynamically generates custom environments based on the specific variables present in the learner's unique assignment. Even when compared to general large language models that generate basic interactive widgets via broad web code, Morphysics stands out by utilizing a specialized architecture. This accurately simulates real-world physics, like friction, collisions, and gravity, inside an easy-to-use, drag-and-drop workspace.



Ultimately, this innovative approach demonstrates clear improvements in the operational business processes of educational institutions, introducing new methods that enhance both teaching efficiency and educational quality. Apparently, the platform eliminates the time-consuming preparation, setup, and teardown of physical lab equipment. This zero-setup model maximizes the utility of a standard 45-minute period, allowing educators to instantly deploy lab environments for an entire class of 40 to 50 students. At an institutional level, it serves as a scalable, cost-effective alternative to maintaining degrading physical infrastructure. Moreover, Morphysics transforms the educational process from passive transcription to active scientific inquiry. By visualizing complex, abstract physical phenomena that exceed the limits of traditional classroom observation, educators can effectively bridge the gap between dry theory and reality. Consequently, this can improve the overall quality of student knowledge comprehension and creativity.

4. BUSINESS MODEL

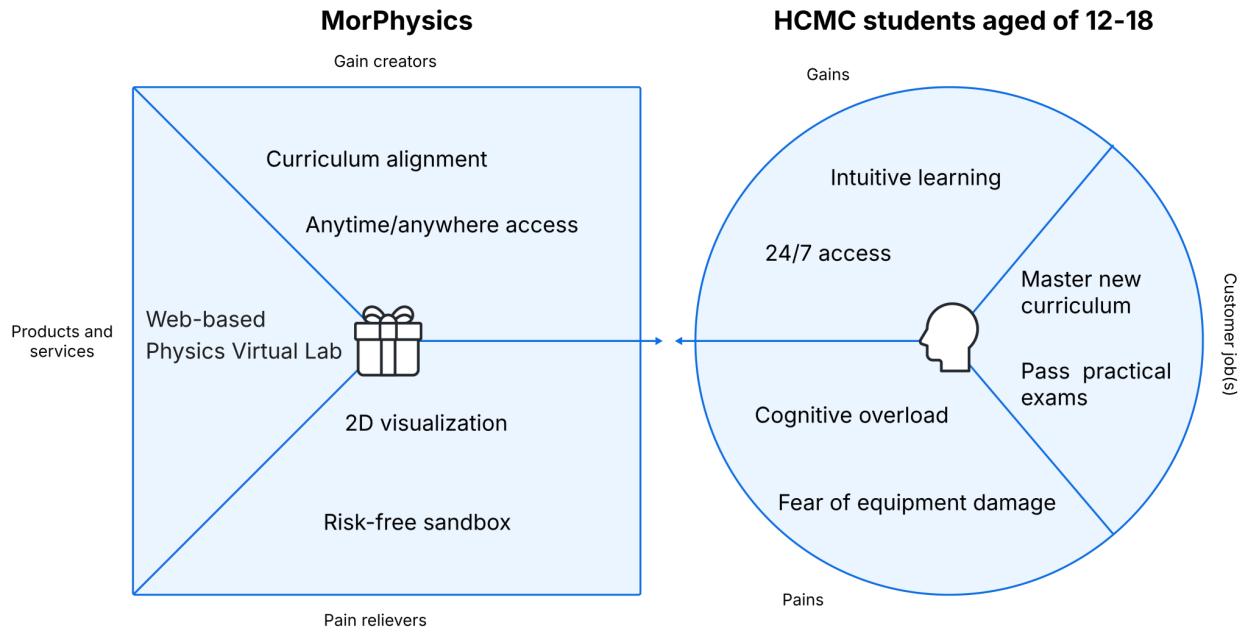


Figure 1. Value proposition map for Ho Chi Minh city students aged of 12-18

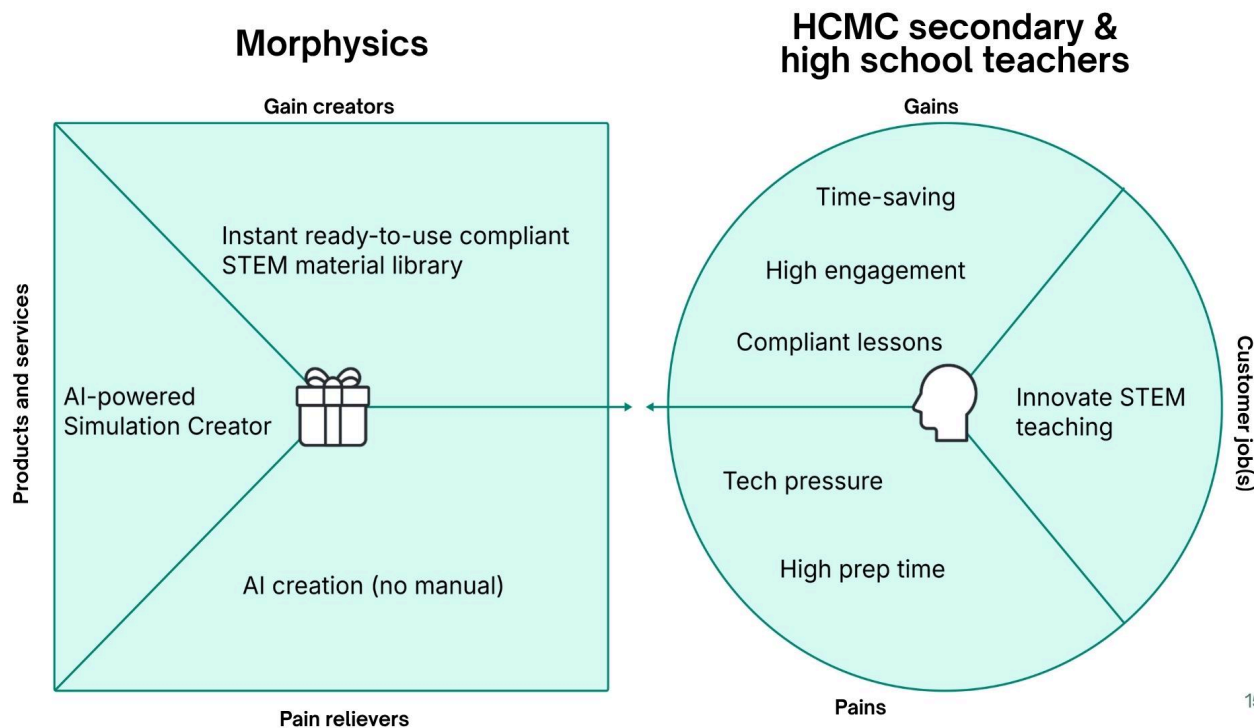


Figure 2. Value proposition map for Ho Chi Minh city secondary and high school teachers

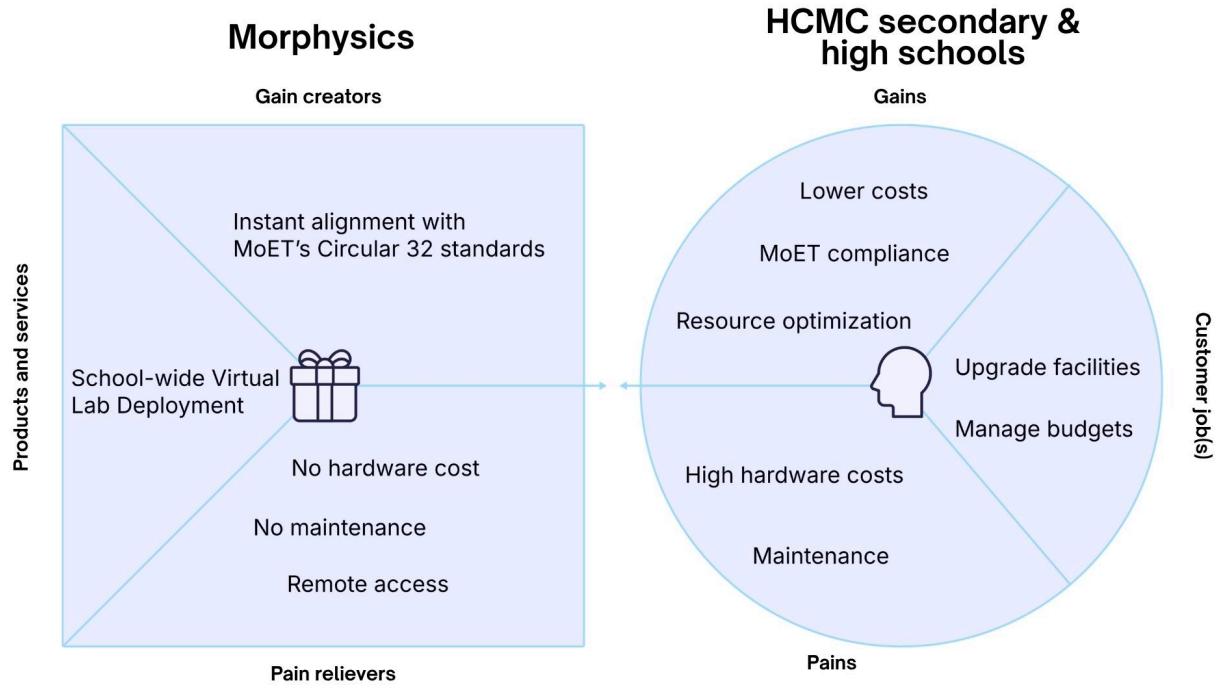


Figure 3. Value proposition map for Ho Chi Minh city secondary and high schools

KEY PATNERS - API providers: Google - Server providers: AWS - Education institutions - Influencers: KOL, KOC - Payment gateways: Momo   	KEY ACTIVITIES - R&D - Software development - Marketing & sales - Customer service KEY RESOURCES - Financial: funding - Infrastructure: server - Human: Development team, R&D team - Intellectual: dataset, system design.	VALUE PROPOSITIONS - Curriculum alignment - Zero-effort customization - Infrastructure-free access - Risk-free exploration	CUSTOMER RELATIONSHIPS Automated self-service: free/basic users Priority service: premium package Dedicated and co-creation: schools CHANNEL - Product-led channel - Educational KOCs - Inbound SEO - Events & media	CUSTOMER SEGMENTS - HCM students aged of 12-18. - HCM teachers in secondary and high schools. - HCM secondary schools and high schools.
COST STRUCTURE Variable costs: AI/LLM API tokens, cloud usage scale, and project-based personnel. Fixed costs: Core server maintenance, marketing/SEO campaigns, permanent salaries, and workspace office.	REVENUE STREAMS - Months 0-6: Launch & Pilot stage - Months 6-18: Validation & Growth stage - Months 18-24: Operational inflection stage - Months 24-30: Full payback & Maturity stage			

Table 1. Lean model canvas for Morphysics

5. MARKET FEASIBILITY

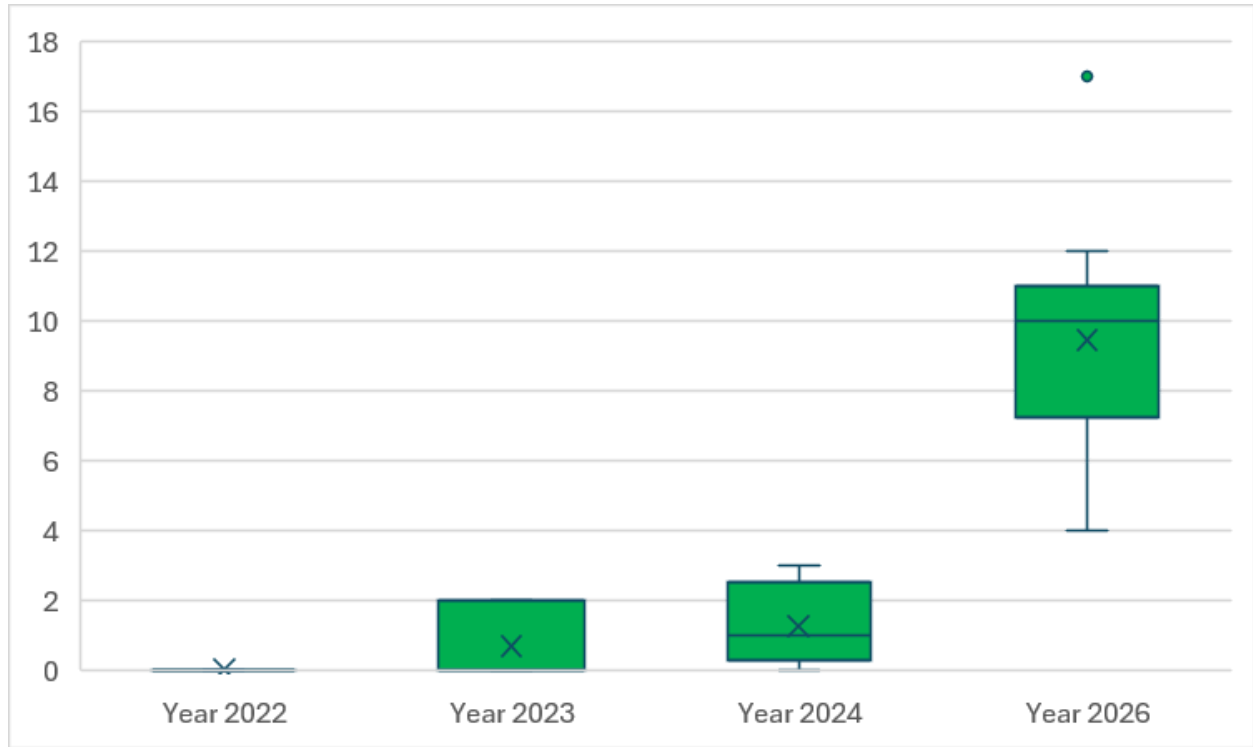


Figure 4. Number of practical questions in Physics examinations



Figure 5. Market segmentation

Assumption.

Number of students per high school : $352.051/181 = 1945$ students (Yen Hoa, 2026)

Number of students per secondary school: $759.278/485 = 1566$ students (Yen Hoa, 2026)

Number of high schools and secondary schools: $485 + 181 = 666$ schools (Yen Hoa, 2026)

The 20,000,000 VND/year budget for the B2B segment is firmly justified by highly optimized unit economics. Operational data indicate that an average high school accommodates 1,945 students (352,051 students / 181 schools), while a secondary school accommodates 1,566 students (759,278 students / 485 schools). Allocating the 20,000,000 VND annual licensing fee across this user base reduces the actual software cost to a marginal range of 10,000 VND to 13,000 VND per student per year. This low cost per user removes financial friction, guaranteeing a high conversion rate in the school procurement process.

The estimated budget for the Educator segment is set at 1,100,000 VND. This figure is derived from the assumption that a user is willing to allocate 5% of their total monthly income. This calculation is based on the income of a newly appointed secondary/high school teacher at Salary Level 1 (coefficient 2.34) in Ho Chi Minh City. A 5% allocation falls well within the acceptable threshold for professional development tools without impacting essential living expenses.

6. INTERNAL ANALYSIS OF THE PROJECT (*)

This app uses AI to read physics problems from text, files, or photos and turns them into visual models, saving students from having to draw the setups. It also features a built-in system that lets users change settings like gravity or friction to see how objects react in real time.

The app faces a few internal challenges. Its accuracy relies a lot on how clear the user's text or photos are. Also, building and running a custom, visual physics system takes much more effort, time, and technical support than creating basic text tools.

There is a growing need for visual learning tools in Vietnam. Since students often find science theory boring, an interactive app is very attractive. It can help millions of students facing heavy exam pressure learn concepts instead of just memorizing facts. Furthermore, teachers can use it to quickly show examples in class without making slides, and parents get an affordable way to help their children truly understand the material.

The app must compete with current study habits and fast-growing AI tools. Many students are already used to getting quick answers from existing math solver apps or using manual simulation websites. Additionally, since big AI models can now create their own interactive tools, this app needs to constantly prove its unique value as a specialized physics space to stay competitive.

7. DEVELOPMENT STRATEGY (*)

System Enhancements:

Phase 1 (Product Upgrade). The primary focus is diversifying data input by implementing Image-to-Simulation and Docs-to-Simulation features. By incorporating new datasets, we create a strong reference base to handle complex and novel problem-solving scenarios. This data foundation allows us to upgrade our core engine, deploying a high-performance LLM as the main intelligence layer to boost the system's reasoning capabilities. Finally, we will add an output generation feature that lets users export their fully rendered simulation canvas directly into a GIF, making it easy to share visual results and naturally increasing the platform's visibility.

Customer reach:

Months 1 - 6 (Market Entry). The primary focus is establishing a Proof of Concept (PoC) by securing the first 3 schools. By pricing the B2B School package at an accessible 7.500.000 VND, we effectively bypass lengthy institutional budget approvals. This direct classroom integration creates a zero-cost network effect, allowing us to organically acquire over 2.000 B2C users as teachers naturally onboard their students.

Months 6 - 24 (Scaling & Cost Control). As we expand from 3 to 16 schools and approach around 17.000 users, fixed costs will predictably rise to support infrastructure upgrades and a full-time operational team. However, variable costs are tightly managed through strict token consumption limits per user. This ensures that server and AI API expenses scale sustainably without causing budget deficits.

Months 24 - 30 (Break-even). This phase marks our critical financial milestone. Driven by the cumulative growth of our user base and institutional partners, the platform will officially erase all accumulated losses and reach the break-even point by month 30.

Post-30 Months (Optimization). Once market habits are firmly entrenched, our strategy pivots from user acquisition to conversion optimization. We plan to gradually increase the B2C paid conversion rate from the conservative 6% baseline to an industry-standard 8-10%. Simultaneously, we will introduce higher-tier Premium B2B packages with advanced analytics to maximize the lifetime value of our existing customer base.



8. HUMAN RESOURCES (*)

Dr. Truong Thi Thai Minh (HCMUT Lecturer) - Project Mentor & Advisor. Leveraging her extensive background in software engineering and architecture, Dr. Truong guides the project's strategic vision and technical direction. She provides expert consultation on technical solutions, evaluates the feasibility of new features, reviews project deliverables, and ensures the product's development aligns with both academic standards and practical viability.

Ton Long Quan - Project Leader & UI/UX Designer & Frontend Developer. Oversees the entire project lifecycle, Responsible for programming the user interface and optimizing the overall user experience, ensuring the application is intuitive, responsive, and user-friendly.

Tran Chanh Hy - Product Manager & AI/Backend Engineer. Monitors quality across all phases, and provides training for team members. Serves as the lead developer for all Artificial Intelligence (AI) features and integrations.

Tran Thanh Loi & Nguyen Huu Nhan - Backend Engineers Co-manage the backend architecture, server-side logic, APIs, and database management. Additionally, they are responsible for technical system documentation, specifically designing Business Process Model and Notation (BPMN) and Entity-Relationship Diagrams (ERD).

Dang Vo Di Luan - Data Analyst, Business & Capacity Planner. Handles market research (surveys) and business strategy formulation. Responsible for system analysis via Use Case Diagrams and leads Capacity Planning. This includes data validation for token consumption, calculating system load capacity for Daily Active Users (DAU), and forecasting resource requirements for various development phases.

Vu The Khuong - Media, Branding & Business Strategy. Directs the project's visual identity, including website color palettes, UI aesthetics, and the design of the official brand mascot. Manages media production and assists in business development strategies.

9. HIGH POTENTIAL TO CONTRIBUTE TO SOCIETY/COMMUNITY

Morphysics has great potential to impact society by making STEM education accessible to everyone. First, it promotes educational equity by providing an affordable, ready-to-use virtual lab. This helps underfunded schools that lack physical equipment, giving all students equal chances to do hands-on experiments.

Second, the platform guarantees safety. Students can learn through trial and error without the physical risks of a traditional lab. Finally, Morphysics turns dry theory into an interactive visual experience. This inspires students and fixes the current drop in science enrollments. Overall, it encourages more students to pursue STEM careers, helping to build a skilled future workforce for Vietnam.

10. THE TEAM'S ASPIRATIONS AND OBJECTIVES IN SUBMITTING THE PROJECT TO THE COMPETITION

Through this competition, our team wants to highlight the importance of practical experiments in STEM education. We believe that combining proper guidance with student passion will help our country's science sector grow. This approach ensures we do not leave any talented students behind.

Many students in remote mountain villages or isolated islands only learn science through dry textbooks. They never get to touch a real magnet, a basic circuit, or simple lab tools because their schools cannot afford them. It is heartbreaking to see their curiosity and passion fade simply due to poverty. We want Morphysics to change this reality. With just a basic internet connection, these disadvantaged children can finally run experiments and see physics come to life. We aim to bring a complete virtual lab to the students who have always been left behind by circumstances.

11. REFERENCES

Category and Cost Estimate of Subject Classroom Equipment (2024, May 6).

https://sotc.langson.gov.vn/upload/105374/20240506/Phu_luc_kem_theo_Thong_bao_lu_a_chon_cac_to_chuc_don_vi_tham_dinh_gia_7a4c8.pdf

Duong, B. T., & Dinh, T. Q. T. (2022). Solutions utilizing smartphones to support 10th-grade students in online Physics experiments. *Can Tho University Journal of Science*, 58 (Special Issue on Education in the Mekong Delta), 154-162.

Ha, T. H. (2024). Difficulties in pedagogical activities of lower secondary school teachers in northern midland and mountainous provinces. *Journal of Science of HNUE: Educational Science*, 69(1), 73-84.

Hoang, C. C., & Nguyen, H. C. (2024). Current status of teaching equipment management in some lower secondary schools in Luc Yen district, Yen Bai province. *Journal of Education*, 24(11), 192-196.

Mathpresso. (2026 May). QANDA ver 7.4.20. <https://qanda.ai>

Nguyen, T. T. T. (2022). Difficulties and solutions to improve the quality of STEM education in teacher training. *Vietnam Journal of Educational Sciences*, 18(2), 46-52.

Tran, T. N. A., Le, T. C., Nguyen, T. N., & Tran, M. N. G. (2021). Using virtual laboratories in teaching Physics: The case of teaching the "Gas" chapter (Physics 10). *Journal of Education*, 509(1), 30-34.

Trần, T. V. A. (2025). Developing practical vocational skills for students through virtual laboratories and digital simulations at Dien Bien College. *Journal of Psychology and Education*, 31(10).

Voice of Vietnam. (2024, February 14). The minimum teaching equipment rate meeting teaching needs reached over 54%. VOV.

VTV24. (2024, May 16). Why are students increasingly avoiding natural sciences? <https://www.youtube.com/watch?v=Bs1ZQKodpew>



University of Colorado Boulder. (n.d.). PhET interactive simulations. <https://phet.colorado.edu>